Assignment 1 - Git & GitHub Fundamentals

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1. Repository Setup

I initialized the project as a local Git repository using git init and connected it to my remote GitHub repository using git remote add origin and configured .gitignore to exclude local logs while keeping important.log tracked

2. README & Commit History

I filled out the Readme.md file with my full name, GitHub username, technologies used, code block , also a link and an image. I did 5 meaningful logical commits across multiple files (README.md, index.html, style.css, and app.js) with descriptive commit messages.

3. Selective Staging

During development, I modified multiple files at once in my workspace. Instead of staging everything together, I used git add <filename> to stage and commit only one specific file first, leaving the other modified file unstaged until a later commit.

4. Branch & Merge

I created a dedicated feature branch named about-feature using git checkout -b about-feature. After making and committing changes on that branch, I switched back to the main branch using git checkout main and successfully merged the feature branch into main using git merge about-feature.

5. Remote Changes

A commit records changes locally, However a push uploads those local commits to the remote repository on GitHub. A git fetch checks the remote repository for updates without altering local files, while git pull fetches and automatically merges those remote changes into the local workspace.

6. Fair Use of AI

I used the AI tool to help explain complex Git commands and clarify error messages. I independently executed, tested, and reviewed every step in my local repository to ensure complete comprehension.

7. Sources & Reflection

Sources: Lab session slides and personal lab notes.

Reflection: The hardest concept was understanding how remote tracking branches interact with local merges when syncing updates. If Git ever behaves unexpectedly in the future, the very first thing I will check is git status to inspect the exact state of my working directory and staging area.